Yiqun ZHANG

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EDUCATION

The Chinese University of Hong Kong (CUHK) Hong Kong SAR
B.Sc. in Computational Data Science *09/2023 – 08/2027 (Expected)*
CGPA: 3.26/4.0 (Major GPA: 3.5/4.0) | Relevant Coursework: Computer Graphics, Algorithms, Machine Learning, Statistics

Dartmouth College (Exchange Program) Hanover, NH, USA
Visiting Student, Computer Science *09/2025 – 12/2025*
Course: AR & VR Development (Grade: A) | Focus: Unity Engine, VR scene construction, animation

TECHNICAL SKILLS

- **Programming**: C++, C#, C, Python, WGSL (WebGPU Shading Language)
- **Graphics & Game Development**: Unity (C# scripting, VR/AR development for Meta Quest 3, animation), WebGPU API, real-time rendering (vertex/fragment shaders, Phong lighting, OBJ mesh loading)
- **Tools & Libraries**: Git, PyTorch, Allure
- **Languages**: Mandarin (Native), English (IELTS 7.5)

PROJECT EXPERIENCE

Unity Game & VR Development Hanover, NH, USA
AR/VR Development Course, Dartmouth College (Exchange) *09/2025 – 12/2025*

- Developed Unity games and XR projects for Meta Quest 3 in a developer role, implementing gameplay mechanics and player interactions in C#.
- Projects include *XR Basketball Arena* (mixed reality, stencil-buffer portal, physical simulation) and *The Thai House* (VR horror game).

WebGPU 3D Rendering Projects Hong Kong SAR
Computer Graphics Course, CUHK *01/2026 – 02/2026*

- Implemented a real-time 3D rendering pipeline with the WebGPU API, writing WGSL vertex/fragment shaders and managing uniform buffers and render passes.
- Applied the Phong lighting model (ambient, diffuse, specular) to render lit 3D scenes with interactive camera/object control.
- Developed a LEGO brick scene builder supporting 30+ brick types with rotation logic and OBJ mesh loading.

RESEARCH EXPERIENCE

Interpretability and Model Fusion for PSViT Physical Property Estimation Hsinchu, Taiwan
Research Intern, National Tsing Hua University *06/2026 – 08/2026*

- Studied an existing PSViT framework for physical mass estimation from multi-view images and point clouds, analyzing its behavior through interpretability experiments.
- Implemented a KMeans feature-space reshape module (K=4) on top of PSViT, reducing ADE by 5.6% over baseline (4.081 vs. 4.322) on the ABO-500 dataset.
- Ran ablation studies comparing spatial clustering (PartField) with feature-space clustering (KMeans) to guide architecture design choices.
- Explored integrating existing models (VGGT, DINO) into the pipeline, with geometry-confidence-aware view weighting to reduce redundant computation.

INTERNSHIP EXPERIENCE

Software Automation Testing Intern Beijing, China
Gangjutong (Beijing) Technology Co., Ltd *07/2025 – 08/2025*

- Developed and executed automated test scripts to improve testing efficiency and product quality.
- Generated test reports with Allure, giving the development team clearer visibility of results and faster debugging.

HONORS & AWARDS

- Yasumoto International Exchange Scholarship, The Chinese University of Hong Kong, 2025
- Talent Development Scholarship, The Chinese University of Hong Kong, 2024
- CDAS Admission Scholarship, The Chinese University of Hong Kong, 2023
- First Prize, Olympic Mathematics Competition in Jiangsu Province, Jiangsu Mathematical Society, 2022